



SpaceX to track missiles

SpaceX has received a hefty contract from the SDA to build missile-tracking satellites. <http://digit.in/nov20-29>



OSIRIS-REx is a chomper

NASA has shared images of the OSIRIS-REx biting onto more than it can chew. <http://digit.in/nov20-30>

Global has launched 32 satellites, which are usually sent up in batches of 4 or 8.

Many of these satellites are of the small CubeSat standard, an increasingly popular platform. The Space-BEEs were even smaller and are considered to be picosatellites. ISRO has its own modular platform developed along the lines of CubeSat, called Indian Nanosatellite Standard (INS). These are satellites that can be built quickly and deployed using off the shelf components. Space technologies are getting increasingly accessible, demonstrated by Canon from Japan that sent up the CE-Sat 1 on the PSLV C38 mission in 2018. It was basically a modded out dSLR (the EOS 5D mk3 if you are curious). All of this goes to demonstrate one thing, there is an incredible demand for launching small satellites at competitive rates, and all the space agencies around the world simply do not have the launch capacity to meet this demand. This is a market that ISRO intends to serve fully.

In 2018, ISRO got approval to go ahead with plans for adding another launch vehicle to its fleet, one dedicated for launching microsatellites and nanosatellites. It also got approval for a dedicated new launchpad to execute these spaceflights. The new launch vehicle was officially christened "Small Satellite Launch Vehicle", in the matter of fact and straightforward fashion of ISRO (remember, this is the same agency that launched a microsatellite called MicroSat). Unofficially or semi-officially, the rocket is referred to as the Baby PSLV. The launch vehicle is designed to take up payloads of up to 500 kg into low Earth orbit, or 300 kg into a Sun synchronous orbit, with a liftoff weight of about 120 tons. The first demonstration flight was supposed to take place in 2019, but that didn't happen, and it was instead rescheduled towards the end of 2020. In January 2020, ISRO chairman Dr K Sivan addressed a press meet at the ISRO HQ in Ben-

galuru and announced plans for a second spaceport in India, to be built in the Thoothukudi district in Tamil Nadu, that would be exclusively used for SSLV flights.

NSIL, which was incorporated on 6 March 2019, has a major role to play in ISRO's plans for future fleet build up. This is because ISRO facilities are not enough to build the large number of SSLVs needed to meet the demand. NSIL is tasked with production of both PSLV and SSLV through partnerships with private industries. Before NSIL, Antrix was given the same task. NSIL will also



The components of the SSLV displayed at the Space Expo 2018 in Bengaluru

Credit: Dean Sumith

be responsible for marketing the launch services, as well as other space services. In June 2019, NSIL held an industry meet, where the plans for getting the private sector involved in the Indian space program was mapped out. Initially, NSIL was looking to source five SSLVs from the private sector. According to tender documents, ISRO is aiming to increase its launches to 10 PSLV flights and 10 SSLV flights every year, a demand which would have to be met by the private Indian aerospace industry. ISRO has indicated that there is sufficient demand for

increasing the number of flights even beyond this number. The components for the rockets would have to be manufactured according to the process laid out by VSSC.

The initial design and development of the SSLV is being done by the Vikram Sarabhai Space Centre (VSSC) at Thiruvananthapuram. The rocket is designed with 3 solid propulsion stages, as well as a velocity trimming module (VTM) which uses liquid propellant. The SSLV is 2 meters in diameter, and measures 34 meters in length. By comparison, the PSLV measures 2.8 meters in diameter, and 44 meters in length.

ISRO typically inaugurates a new launch vehicle with a series of demonstration flights, which are not considered official missions. The first satellite to be deployed is most likely to be the Microsat-2A, which is meant to demonstrate the launch on demand capabilities of the SSLV. For the second demonstration flight, NSIL has already bagged a private contract, in the form of four satellites for BlackSky global.

Over the past year, ISRO has been taking the necessary steps to prepare the SSLV for its first demonstration flight. This includes static firing of the first stage motor, testing of the second stage flight motor case, with the third stage going through propellant processing. On board is a new Mission Management Computer (MMC) as well as Sequence Execution Module, which together ensure that the flight is as expected, the separation of stages go according to the mission profile, and the satellites are deployed at the right altitudes. ISRO has also managed to reduce the mass of the telemetry package on board by 70 percent.

The real value offered by the new SSLV rocket though, is the fact that students around the world can design satellites in their technical institutions, and get first hand experience on working with instruments in space. Five launches have already been made by Indian students. 🇮🇳